Angular App
Is Slow? 🐌

Here's how I improve performance & speed in real projects 🚀

→

First Step: Analyze

Before optimising, analyse first 🔍

- Identify slow pages
- Check unnecessary re-renders
- Measure load time & TTI

Tools I use:

- Chrome DevTools
- Lighthouse
- Angular DevTools

“No guessing. Fix what actually slows the app.”

Change Detection (Biggest Win ⚡)

Default Change Detection = Performance killer for large apps

- Angular checks the entire component tree
- Even for small changes 😵

✅ Solution:
ChangeDetectionStrategy.OnPush
✔ Updates view only when needed
✔ Huge performance boost in large apps

Lazy Loading

By default, Angular uses Eager Loading 🚨

- Entire app loads at once
- Slower initial load

✅ Use Lazy Loading

- Load modules/components only when user navigates
- Faster startup
- Better user experience

Optimise Lists with trackBy

Rendering large lists? This matters 👇

❌ Without trackBy

- Full list re-renders on change

✅ With trackBy

- Only changed items update
- Less DOM manipulation
- Faster UI

Memory Leaks & Observables

Unmanaged subscriptions = hidden performance issues 🧠

Best practices:

- Use async pipe
- Unsubscribe in ngOnDestroy
- Use takeUntil

✓ Prevent memory leaks

✓ Improve long-term app performance

Assets & Template Optimization

Small things make a big impact ✨

- Lazy load images (loading="lazy")
- Optimise image size & format
- Avoid heavy logic in templates
- Use pure pipes
- Avoid calling functions in HTML

Final Checklist

Angular Performance Checklist ✅

✓ OnPush Change Detection
✓ Lazy Loading
✓ trackBy in lists
✓ Proper subscription handling
✓ Image & asset optimization
✓ Measure using performance tools

Performance is not optional.
It’s part of good Angular architecture. 💯

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